Weizheng Wang

Computer Vision · Multimodal Learning · UAV/Edge AI

Adelaide, Australia | Available in China from Sep. 2026

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Research Profile

Master of Artificial Intelligence and Machine Learning graduate, with a Computer Science background and research spanning **multimodal UAV decision evaluation, visual memory, efficient computer vision and edge deployment**. Co-first-author work covers protocol-conditioned benchmarks, maritime small-object detection, structured pruning and detector feature analysis. Engineering experience includes MindSpore model integration, video analytics, Jetson inference, and open-source developer and desktop tools.

Education

Adelaide University — *Adelaide, Australia*

Master of Artificial Intelligence and Machine Learning — *Sep. 2025 – Aug. 2026*

- Completed 18 August 2026; degree conferred 7 October 2026. Testamur issued by Adelaide University.
- Studied at the University of Adelaide with research interests in computer vision, multimodal learning and NLP for UAV and edge intelligence.
- Recipient of the 2025 University of Adelaide Global Citizens 30% International Scholarship.

North China Institute of Science and Technology — *Langfang, China*

Bachelor's Degree in Computer Science and Technology — *Sep. 2021 - Jul. 2025*

- GPA: 4.02.
- Coursework and research focused on deep learning, computer vision, and UAV-based edge deployment.

Research Experience

Zayed University — *Remote*

Research Assistant – MulRobBench: Safe Multimodal UAV Agents — *Feb. 2026 – Aug. 2026*

- Co-defined protocol-conditioned UAV decision tasks and designed a 3,024-sample benchmark spanning 17 task-taxonomy nodes under visual degradation and noisy operator instructions.
- Built end-to-end data generation, inference and evaluation pipelines for 17 multimodal models, with semantic and structural diagnostics for safety-policy compliance.
- Co-first-authored the resulting manuscript, submitted to Applied Soft Computing; released as arXiv:2607.23870.

Institute of Software, Chinese Academy of Sciences — *Remote*

Research Intern – NLP & Vision-Language Model Fine-tuning — *Feb. 2025 – Jun. 2025*

- Contributed to the MindNLP open-source project by implementing and fine-tuning models such as Autoformer, BEiT and ALBERT within the MindSpore framework.
- Focused on model integration, training stability and cross-task evaluation across NLP and vision-language tasks, improving reproducibility and robustness of training pipelines.
- Collaborated with open-source mentors and contributors through code reviews, issues and documentation.

Beijing Elite Intelligence Company & NCIST — *Beijing & Langfang, China*

Research Intern – Automatic Inspection and Rescue Based on Drone Nest — *Jan. 2024 – Jul. 2024*

- Built drone-based automatic inspection services in open-pit mines, including water count detection, conveyor belt deviation detection and truck tracking using DeepSORT + YOLOv8.
- Optimized PyTorch detection models and exported them via ONNX for real-time inference on edge devices in long-duration UAV missions.

Hebei Province Key Sci. & Tech. R&D Project (No. 19270318D) — *Hebei, China*

Undergraduate Researcher – Baiyangdian Ecological IoT 3D Monitoring — *Jun. 2023 – Jan. 2024*

- Developed object detection for mining trucks using YOLOv7-tiny (Darknet), improving recognition accuracy by 2% with minimal FPS loss.
- Enhanced dual-modal (infrared + visible) UAV object detection with YOLOv8 for low-altitude drone monitoring, achieving 65% mAP at 30 FPS on a laptop and targeting deployment on edge devices.

Preprints & Submitted Manuscripts

†Equal contribution / co-first authors. *Corresponding author.

MulRobBench: A Decision-Level Benchmark for Safe and Security-Policy-Compliant Multimodal UAV Agents

Belal S. Alsinglawi†, **Weizheng Wang**†, Junyi Wu†, Lianhai Lin, Mérouane Debbah, Izzat Alsmadi

2026 | Submitted to Applied Soft Computing | arXiv:2607.23870

Co-developed an offline, protocol-conditioned UAV decision benchmark with 3,024 samples, 17 task-taxonomy nodes and 12 scoring dimensions; evaluated 17 multimodal models. Separates semantic decision quality from structural compliance, unsafe actions and parsing failures.

H2T-DEIM: Hilbert–Haar Representation with Tri-Modal Multi-Dimensional Pruning for Maritime Small-Object Detection from Uncrewed Aerial Vehicle Imagery

Junyi Wu†, **Weizheng Wang**†, Liqin Tian, Qing Tian, Yong Niu, Yi Jiang

2026 | Submitted to Remote Sensing

Combines Hilbert-Mamba context modeling, Haar wavelet downsampling and structured pruning across CNN, Transformer and state-space widths. H2T-DEIM-M achieves 51.1% AP on SeaDronesSee and 64.8% on AFO, with 32.1 FPS on Jetson Orin Nano 8GB under the manuscript's end-to-end protocol; contributed methodology, software and validation.

A Multi-Source Data Fusion and PSO-Transformer Decision Support System for Gold Price Volatility Forecasting and Risk Warning

Xueping Diao*, Xueqiang Lv, **Weizheng Wang**, Junyi Wu

2026 | Mathematics (MDPI) | Under revision

Combines structured market data with TFN-based video sentiment and Qwen-2.5 news sentiment, using PSO-tuned Transformer forecasting and three-tier risk warnings. Reports 74.5% risk-tier classification accuracy in a 102-trading-day backtest.

TRIM: Dual-Budget Structured Pruning with Information-Modulated Saliency for CNN, Transformer, and Mamba Vision Models

Weizheng Wang†, Junyi Wu†, Qing Tian, Liqin Tian

2026 | Submitted to IEEE ICASSP 2027 | [Repository]

Uses dependency-closed pruning units, information-modulated Taylor/Fisher saliency, coupled MILP allocation and measured-budget repair. On ImageNet-1K, Top-1 losses are 0.52/0.66/0.58 percentage points for ResNet-50/DeiT-Base/VMamba-V2 Tiny, retaining at most 69.5% of parameters and 69.4% of forward GFLOPs with zero measured budget violation.

When Retrieval Is Not Binding: A Diagnostic Benchmark for Visual Episodic Memory

Weizheng Wang†, Junyi Wu†, Liqin Tian, Qing Tian

2026 | Submitted to IEEE ICASSP 2027 | [Repository]

Designed matched memory-off, wrong-entity and shuffled-binding interventions while fixing queries, candidates, features and memory envelopes. The development cohort contains 48 objects, 384 observations and 192 queries; on 128 memory-visible queries, three of four training-free readers are unchanged by shuffled binding, separating content benefit from episodic binding sensitivity.

ManiSeg: A Semantic-Entropy Driven Manifold Collision Framework for High-Dimensional Feature Evaluation of Detectors

Junyi Wu†, Zitong Shen†, **Weizheng Wang**†, Xiaoan Shi, Yi Jiang, Liqin Tian

2026 | Submitted to IEEE ICASSP 2027 | [Code]

Co-developed semantic-entropy-driven granulation, geometric manifold-collision-based adaptive three-way decision and a composite Manifold-Semantic Fidelity metric; evaluated detector feature representations on COCO across multiple model variants.

EmergUAV-Bench: Protocol-Conditioned Evaluation of Vision-Language-Action Models for Emergency UAV Decisions

Junyi Wu†, **Weizheng Wang**†, Yi Jiang†, Liqin Tian

2026 | Submitted to IEEE ICASSP 2027 | [Code]

Co-developed a 1,280-sample benchmark combining RGB, LiDAR and pose observations with six language conditions, six core emergency tasks and 16 decision facets. Evaluated 13 multimodal models; the best protocol score is 0.4165, with distinct leaders in protocol agreement, safe-action accuracy and unsafe-action avoidance.

Peer-Reviewed Publications

A Lightweight Thermal Denoising and Occlusion-Robust Infrared Detection Model for Substation Equipment — *IEEE ICIP*

Junyi Wu†, **Weizheng Wang**†, Yong Niu, Xiaoan Shi, Liqin Tian — *2026*

- PO-YOLO achieved 83.5% mAP50 and 56.9% mAP50:95 on ISED-1K, improving the YOLOv11 baseline by 3.4 and 2.0 points while reducing computation by 6.35%. †Equal contribution. [Code]

A lightweight insulator defect detection algorithm based on drone images for power line inspection — *Engineering Research Express 7, 045355*

Junyi Wu, **Weizheng Wang**†, Wanyan Du†, Zhi Liu, Lianhai Lin, Yuchen Cui, Chong Wang, Liqin Tian — *2025*

- LFL-YOLO achieved 97.4% accuracy while reducing parameters by 23.5% and computation by 9.6% versus YOLOv8n. †Equal contribution. [Paper]

Real-time monitoring of trucks used in open pit based on aerial video of UAV — *IEEE RAIIC, pp. 130–138*

Weizheng Wang, Junyi Wu, Yongfei Li — *2024*

- Developed a real-time UAV-based monitoring system for open-pit mine trucks using aerial video. [Paper]

Research on industrial meter readings based on improved YOLOv8 — *Proc. SPIE 13181, 131815G*

Junyi Wu, **Weizheng Wang**, Lei Yang, Xuesong Ni, Liqin Tian — *2024*

- Proposed an improved YOLOv8 approach for robust industrial meter reading. [Paper]

Honors & Leadership

2025	**Recipient,** University of Adelaide Global Citizens 30% International Scholarship	*Australia*
2024	**Principal Investigator,** National Undergraduate Innovation Training Program (202411104021)	*China*
2023	**Grand Award,** National 3D Digitization Innovation Competition	*China*
2023	**Second Prize,** Challenge Cup Science and Technology Invention and Creation Track	*China*
2023	**First Prize,** National College Students' Data Analysis Competition	*China*

Patents

A Dual-Optical Personnel Search and Rescue Positioning Visual System on Drones — *China National Intellectual Property Administration*

Co-inventor — *Nov. 2024 - Present*

- Invention patent under substantive examination. Publication No. CN 119429233 A.

Automatic Battery Change Arm for Drones — *China National Intellectual Property Administration*

Co-inventor — *Dec. 2023*

- Utility model patent granted. Patent No. ZL 2023 2 3426097.6.

Research Projects

National Undergraduate Innovation and Entrepreneurship Training

Project lead: 202411104021 (2024); participant in collaborating student projects

Collaborating projects: 202511104021, 202511104025, 202611104287

Undergraduate research in UAV perception, automated inspection and edge inference. Collaborating projects are led by Yuxia Gao, Lucan Tang and Yi Jiang, respectively.

National Key R&D: Multimodal Perception and Emergency Decision Research

Research participant | Maritime detection, model compression, visual memory and UAV decision evaluation

2024YFC3016800; 2024YFC3016805

Contributed to H2T-DEIM, TRIM, ManiSeg, the visual episodic-memory diagnostic and EmergUAV-Bench. Work spans model implementation, evaluation pipelines, feature analysis and protocol-conditioned multimodal benchmarking.

Intelligent Mine Inspection and Monitoring

Research participant | UAV video analytics and industrial inspection

2022YFC3004702-4; HNCTR23-05

Participated in research associated with coal-mine disaster monitoring and digital decision systems, and long-distance coal-flow roller/belt monitoring. Related undergraduate work includes mining-truck detection/tracking, standing-water counting and conveyor-belt deviation detection.

Multimodal Airborne Search-and-Rescue Equipment

Research participant | Equipment development and demonstration | Project budget: RMB 800,000

Related regional programs: 202622003010002; 25350802D; 2025011099

Related work includes infrared-visible UAV detection, personnel search/localization and edge inference. Regional participation covers Hebei major science and technology support, Hebei innovation-capacity enhancement and Langfang science and technology support programs.

IoT Research Platforms and Collaborative Research Funds

Research participant | Sensing, visual detection and research software

21567693H; 2017-ZJ-Y21; 3142024038; 3142026024; 262Y5403D

Hebei IoT Monitoring Technology Innovation Center; Qinghai IoT Key Laboratory; Fundamental Research Funds for the Central Universities; Hebei collaborative innovation special project on livelihood security and social safety. The undergraduate Baiyangdian project (19270318D) is detailed in Research Experience.

Selected Open-Source Projects

- **Jetson VLM Lab**: GGUF vision-language model deployment and text/image smoke evaluation on Jetson using llama.cpp and Docker.
- **DEIM Jetson Acceleration**: ONNX/TensorRT deployment and FP16/INT8 benchmarking with correctness, latency and power measurements.
- **Video-based Intelligent Alert System**: Undergraduate thesis: multi-stream video monitoring, open-vocabulary detection, alerts and multimodal Q&A.
- **DailyPaper**: Automated AI paper collection with a searchable, categorized web archive.
- **CleanSlateTab**: Browser new-tab extension with configurable search, shortcuts, wallpapers and WebDAV sync.
- **SnapPin**: C++ Windows utility for screenshots, annotation, OCR and pinned images.

Technical Skills

Languages	Python; C++, JavaScript/TypeScript, Java; PowerShell; SQL; LaTeX
ML & Evaluation	PyTorch, MindSpore/MindNLP, Transformers; multimodal benchmarks, visual memory, structured pruning
Vision	YOLO/DEIM, OpenCV, DeepSORT; thermal-visible perception, feature analysis, granular computing
Edge & Deployment	ONNX, TensorRT, llama.cpp/GGUF, CUDA, Jetson, Docker, Linux/WSL
Applications & Tools	Django, Flask, SQLite/MySQL, ZLMediaKit/FFmpeg; browser extensions; Git, CMake, GitHub Pages